

A low-angle, upward-looking photograph of several tall, fluted classical columns. The columns are made of light-colored stone or marble and are topped with ornate capitals. They converge towards the top of the frame, creating a strong sense of height and grandeur. The background is a clear, bright blue sky. The lighting suggests it's daytime, with some shadows visible on the columns.

Capítulo 14

Finalizando



PERF

vs

SCALE

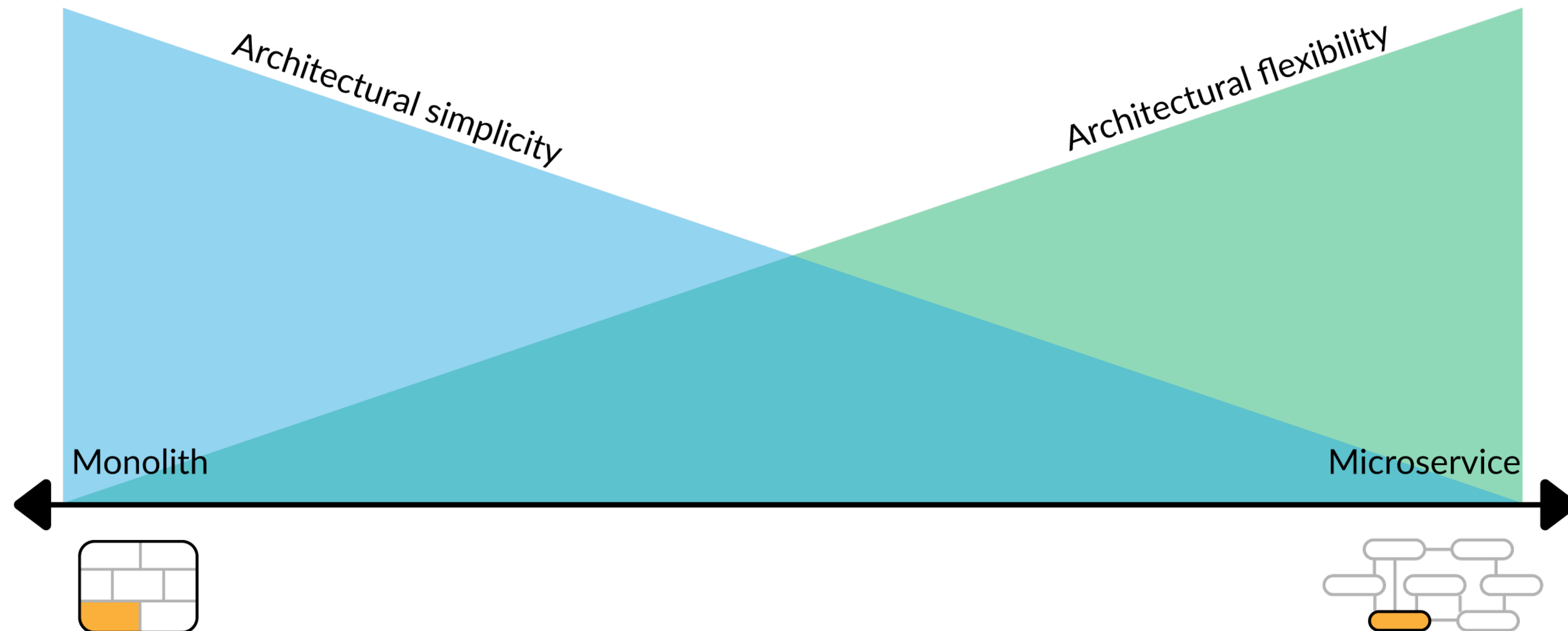


Monolith

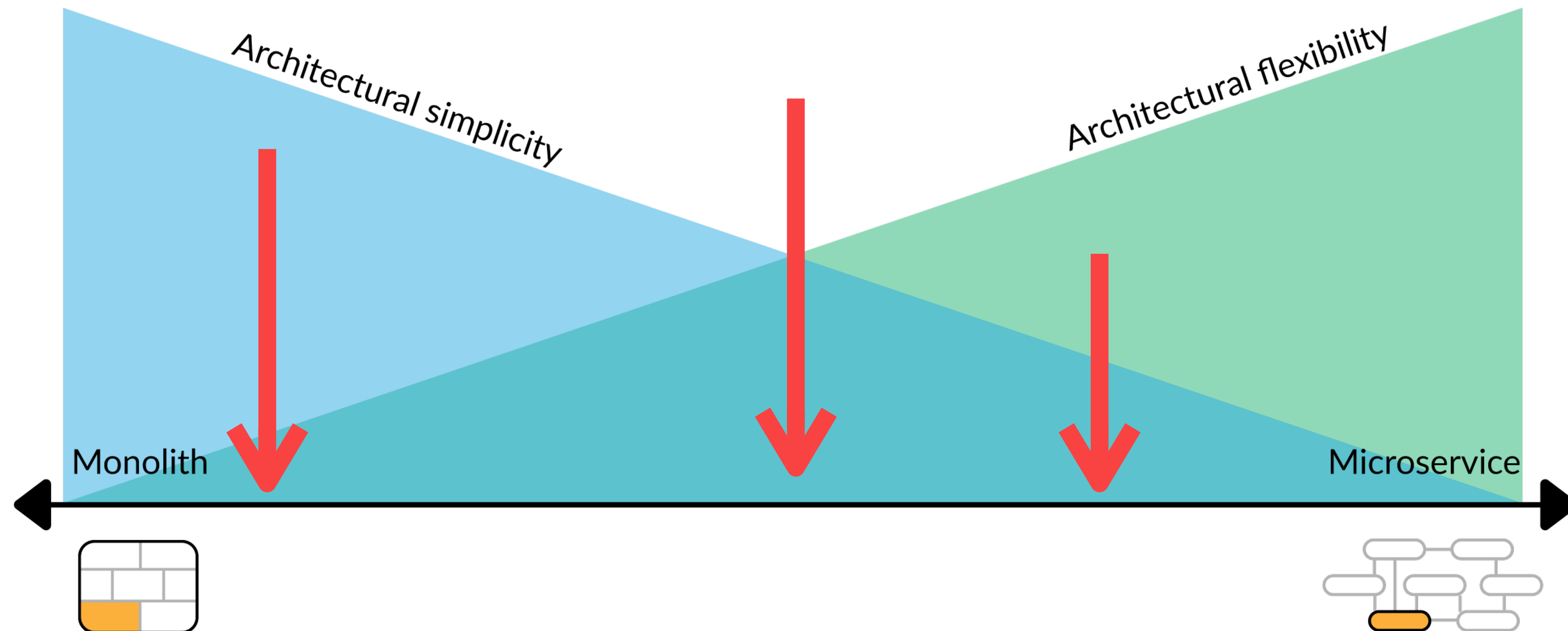


Microservices

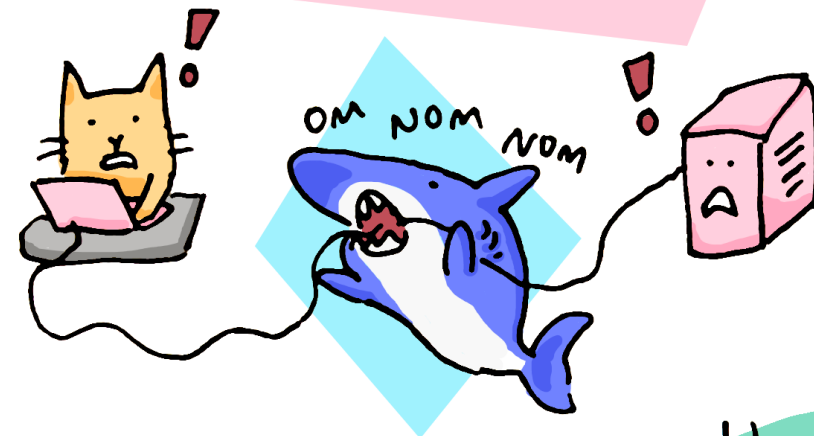
The monolith to microservices SPECTRUM



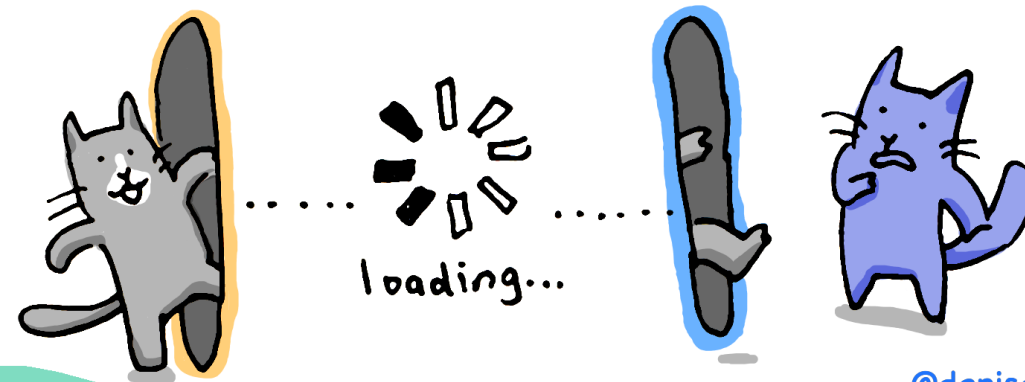
The monolith to microservices SPECTRUM



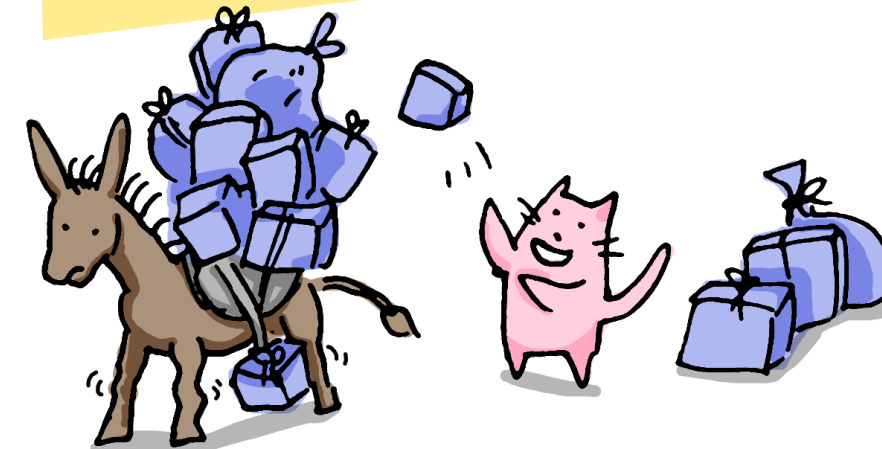
① The network is reliable



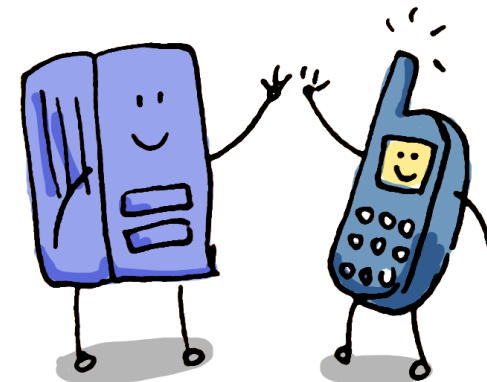
② Latency is ZERO



③ Bandwidth is infinite



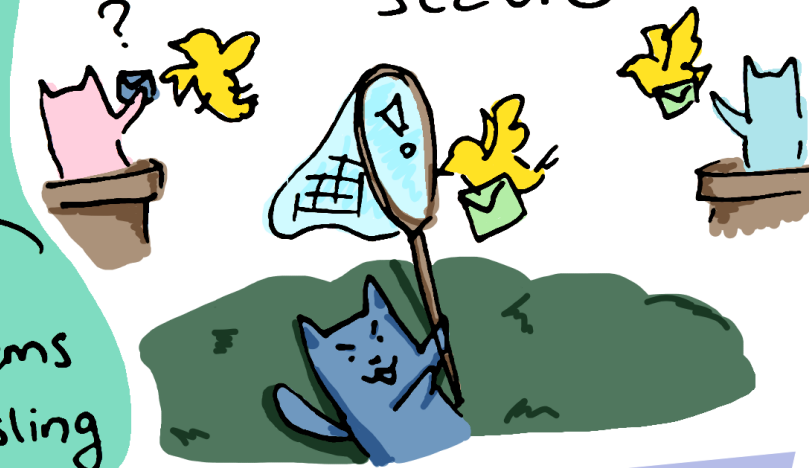
⑧ The network is homogeneous



the 8 Fallacies of Distributed Computing

Originally formulated by L. Peter Deutsch & colleagues at Sun Microsystems in 1994; #8 added in 1997 by James Gosling

④ The network is secure



⑦ Transport costs \$0



⑥ There is only one administrator



⑤ Topology doesn't change



**Arquitetura é sobre
contextos, requisitos e
restrições**



Enterprise





Enterprise



Startup



Enterprise



Startup



Big Tech

SCALE

FUNDAMENTALS



LOW
LATENCY



LOW
LATENCY



THROUGHPUT
HIGH

**LOW
LATENCY**



**THROUGHPUT
HIGH**

**AVOID
COORDINATION**

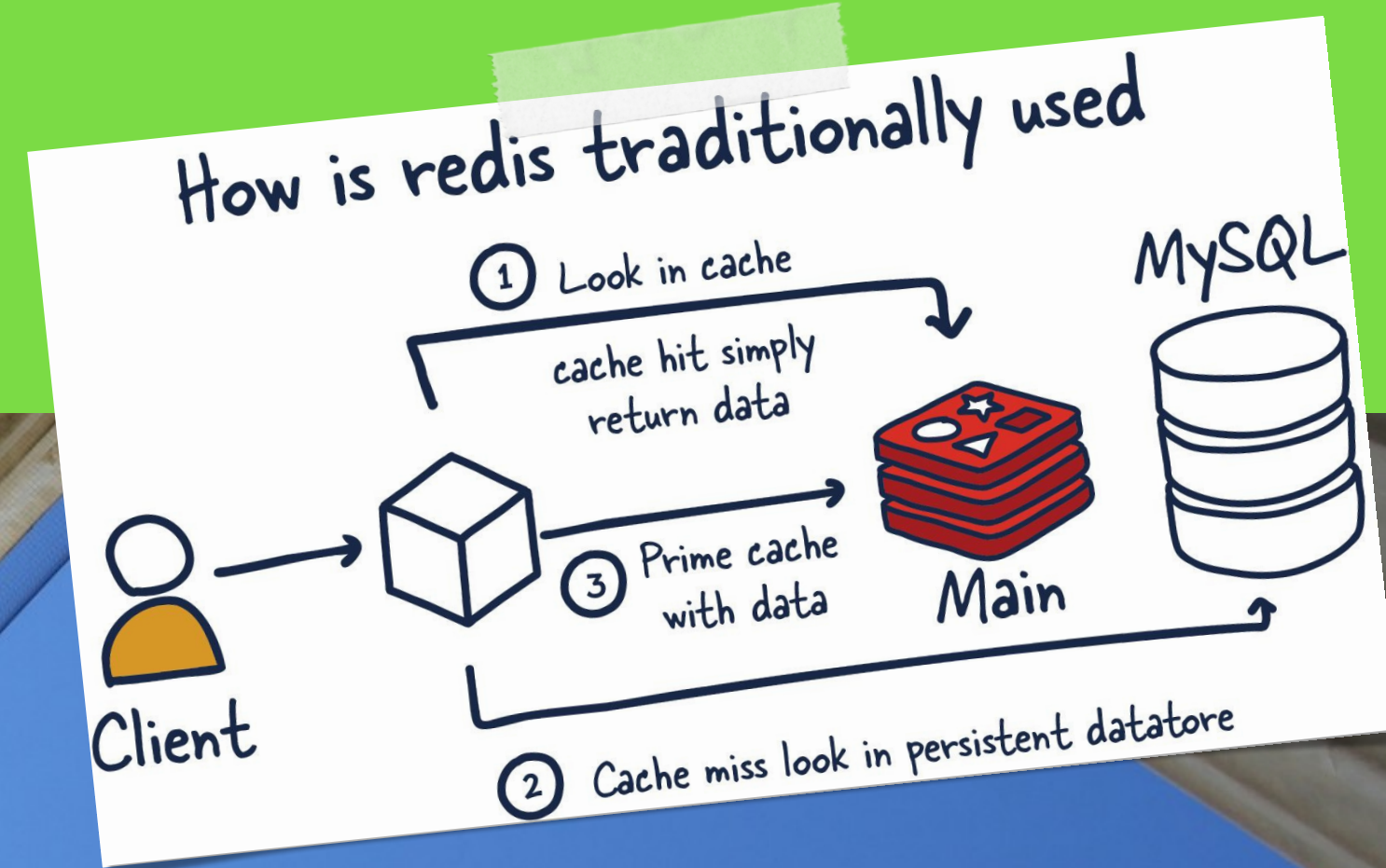




CACHE



CACHE

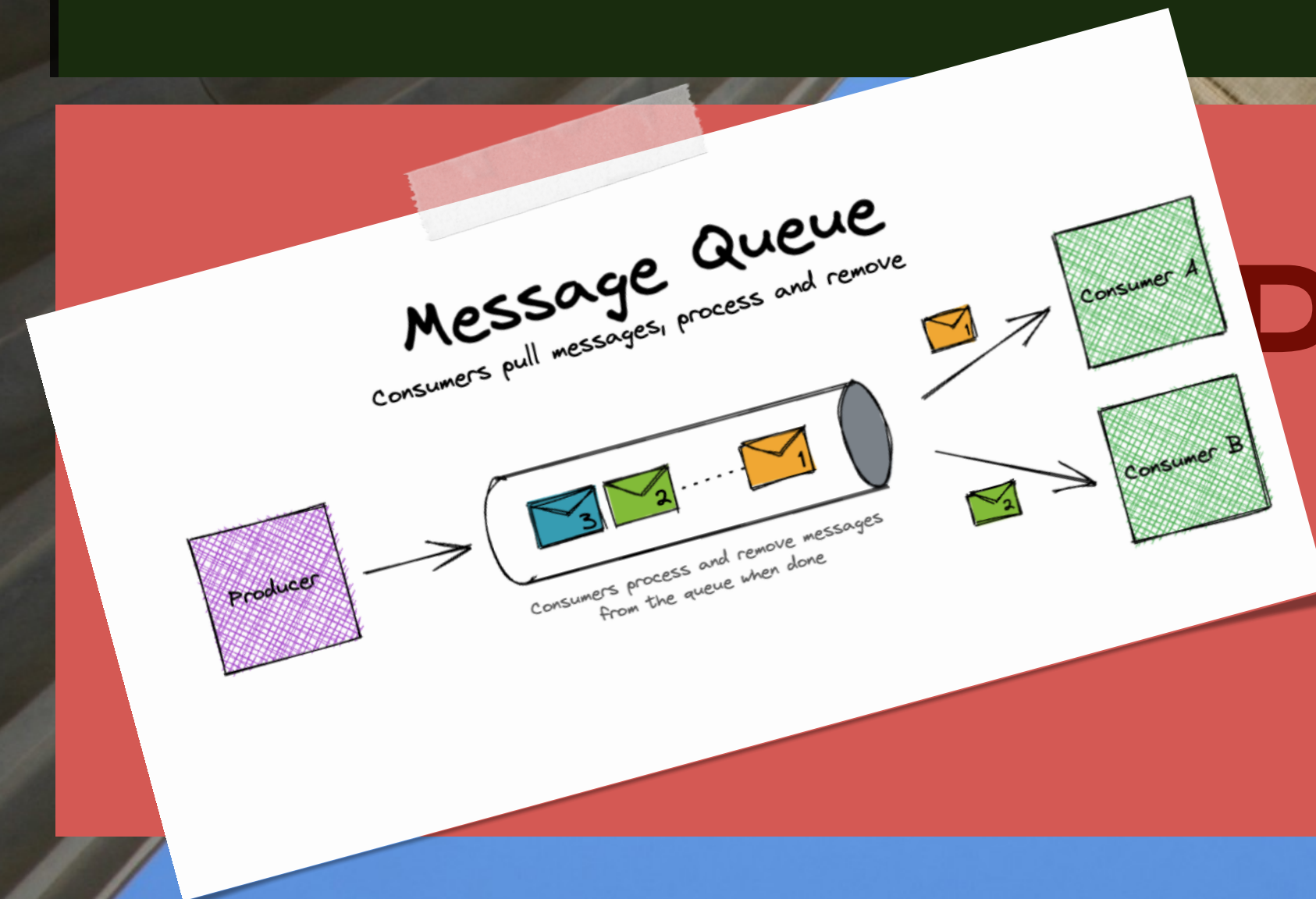




CACHE

PROCESSAMENTO
ASSINCRONO

CACHE



PROCESSAMENTO ASSINCRONO

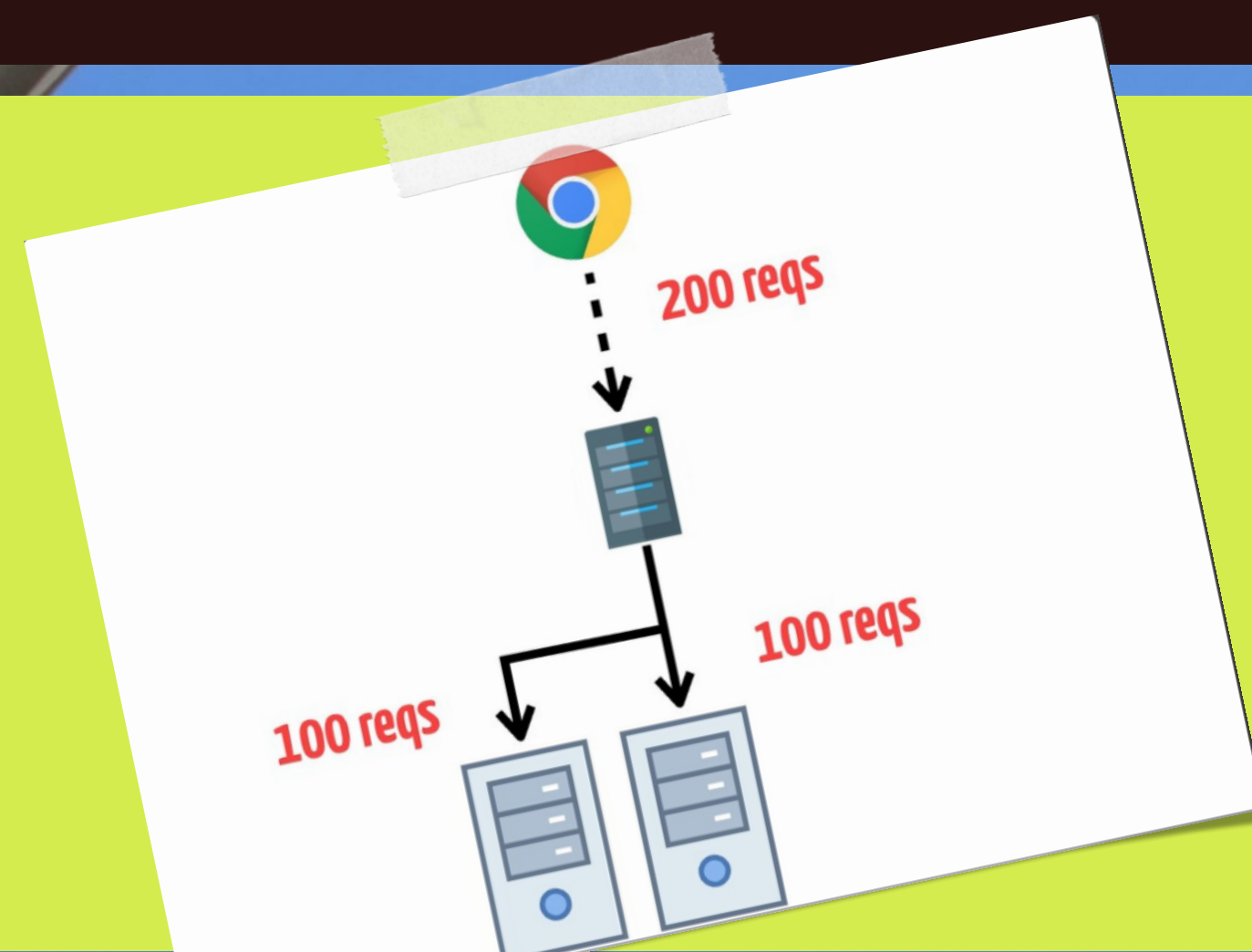
CACHE

PROCESSAMENTO
ASSINCRONO

BALANCEAMENTO
DE CARGA

CACHE

PROCESSAMENTO ASSINCRONO



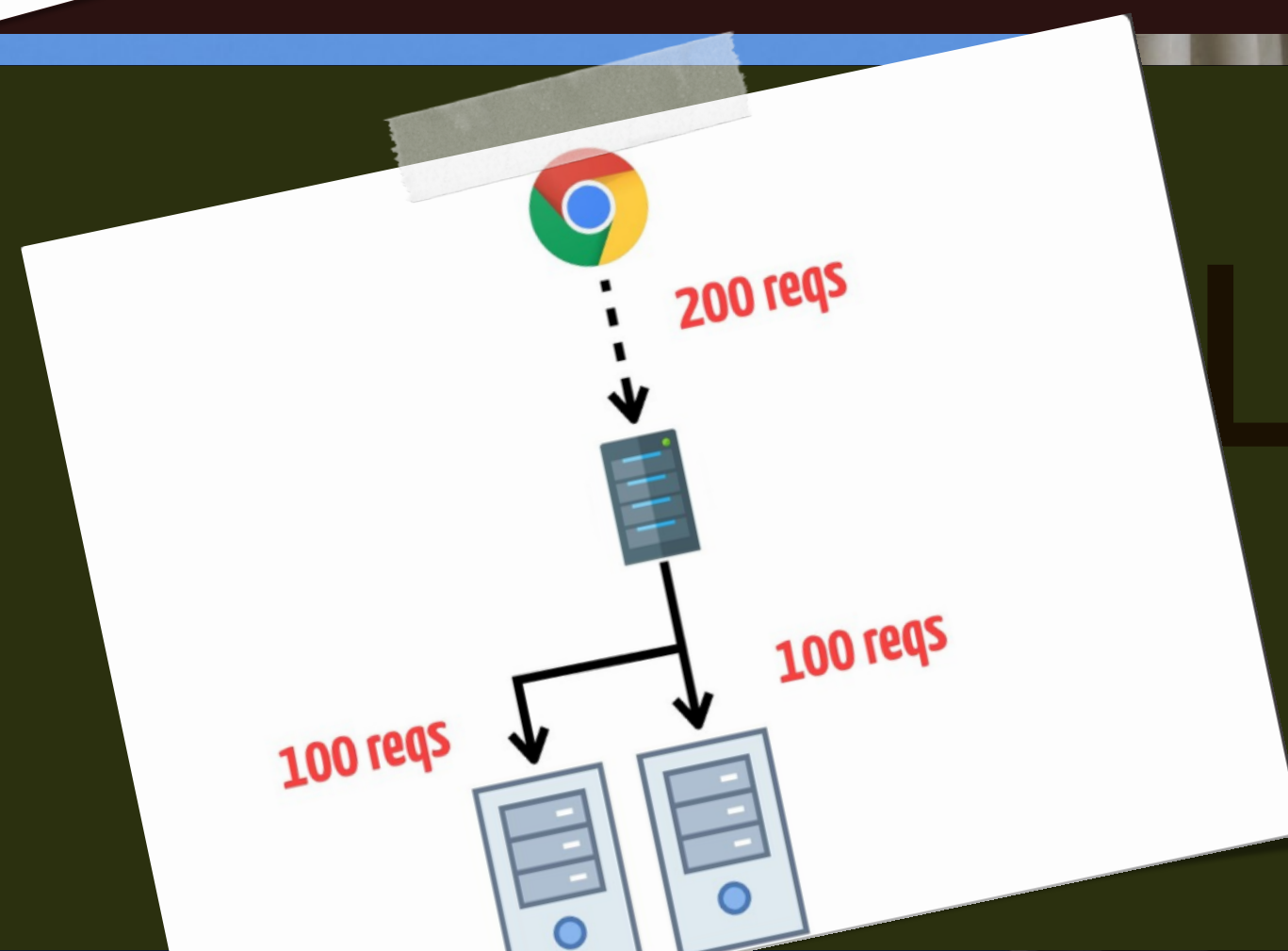
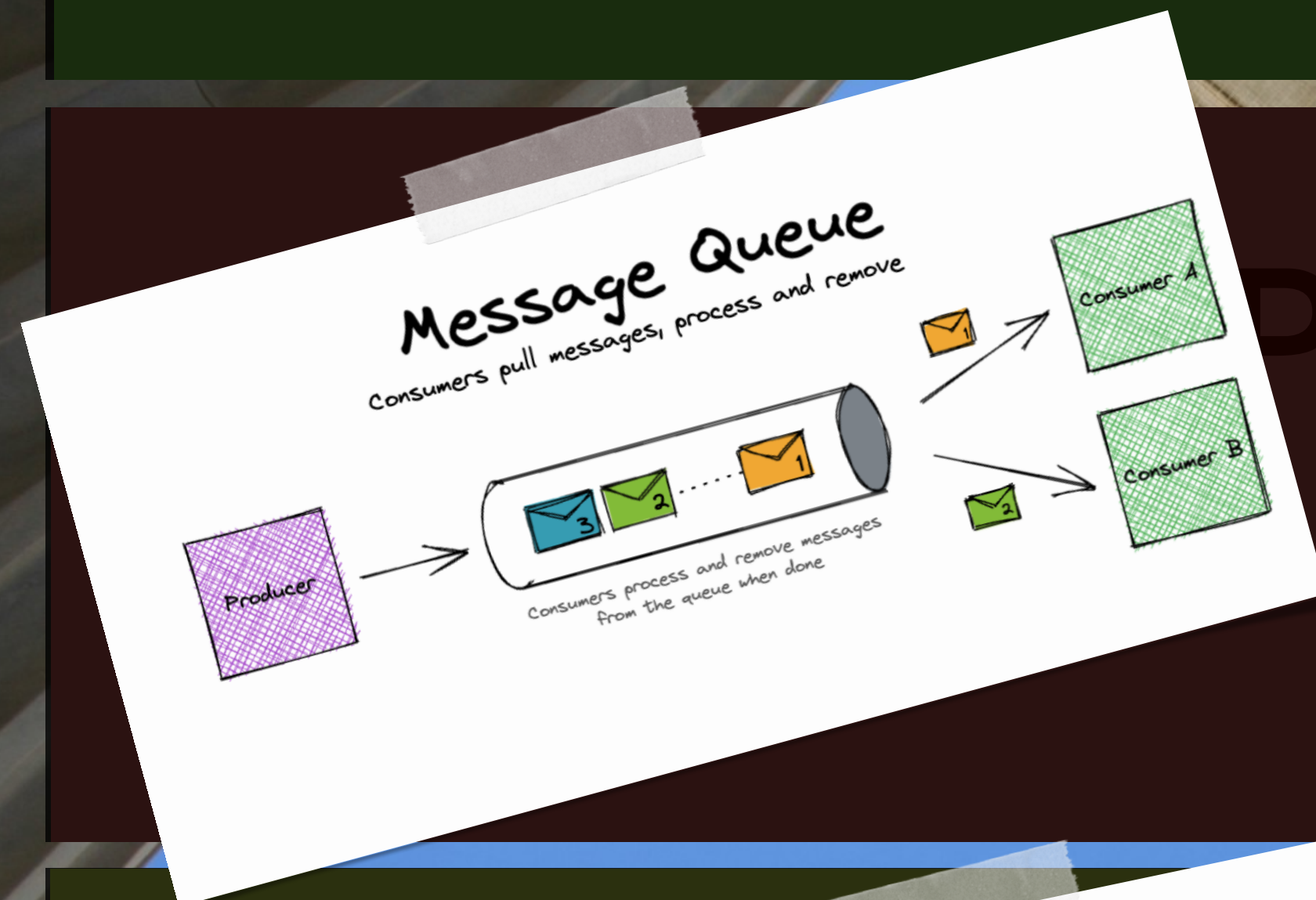
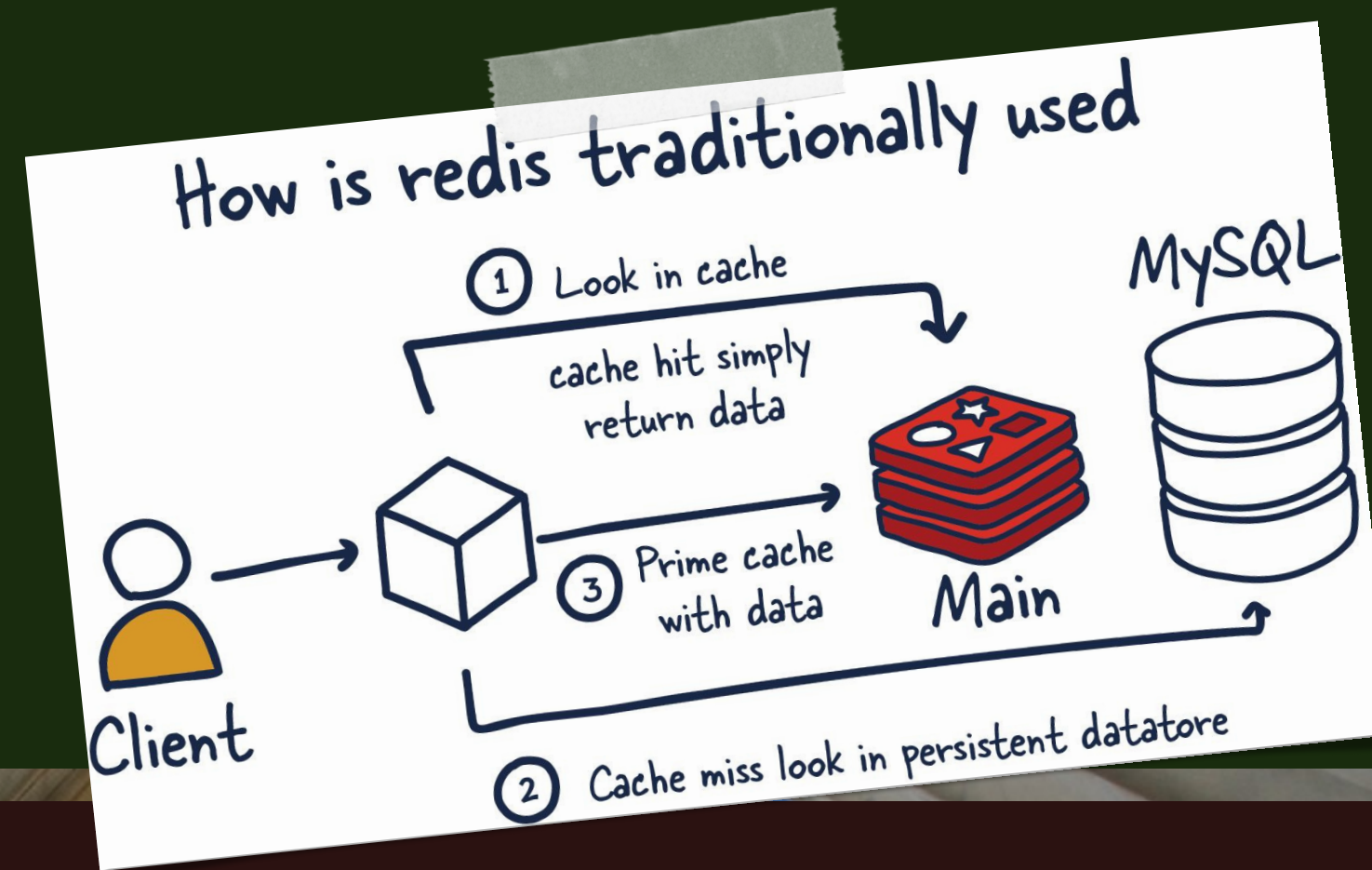
BALANCEAMENTO DE CARGA

CACHE

PROCESSAMENTO
ASSINCRONO

BALANCEAMENTO
DE CARGA

CACHE



PROCESSAMENTO ASSINCRONO

LANCEAMENTO DE CARGA



trade-off

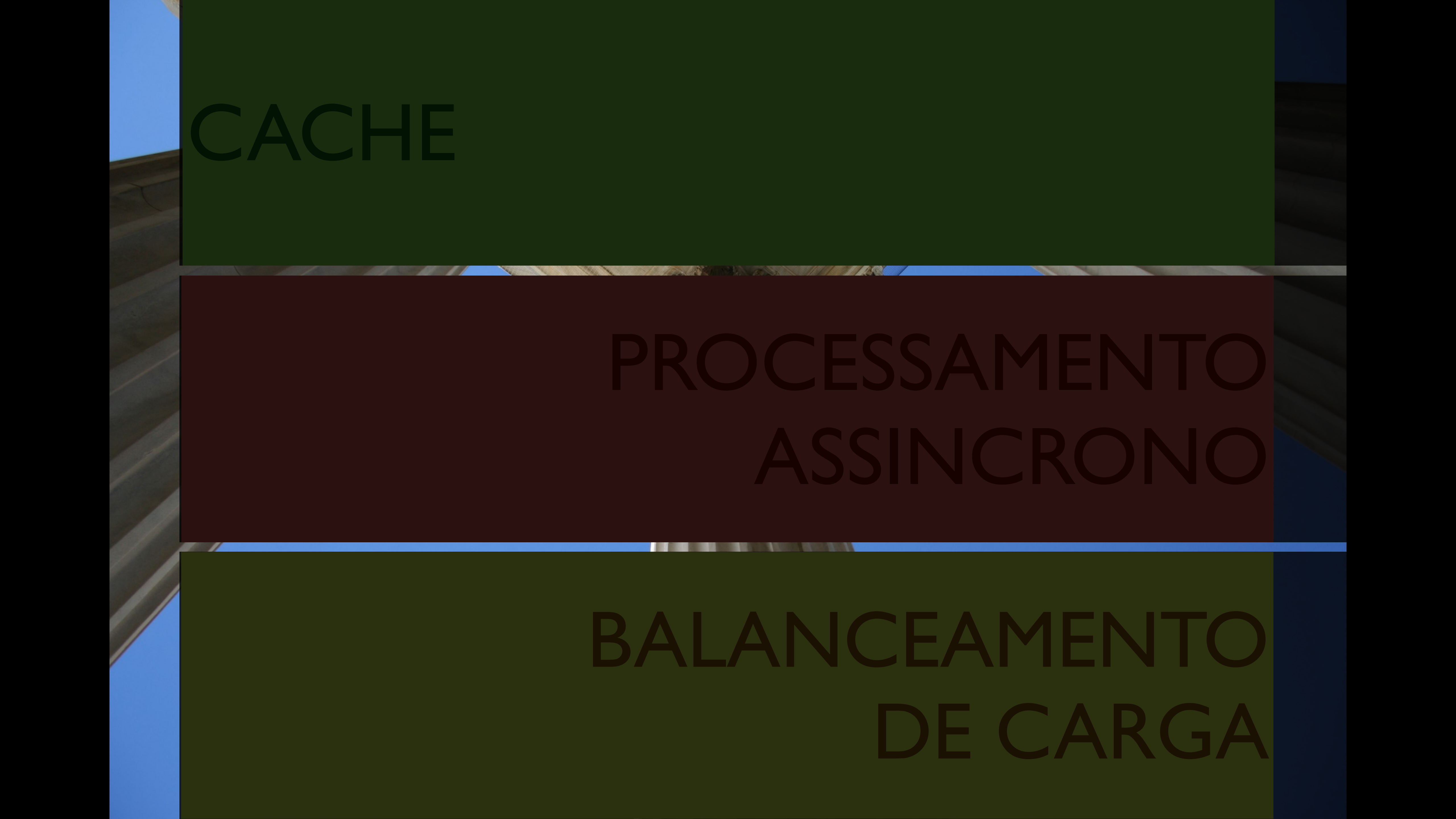
/ˈtrād,ôf/

noun

noun: **tradeoff**

a balance achieved between two desirable but incompatible features; a compromise.

"a trade-off between objectivity and relevance"



CACHE

PROCESSAMENTO
ASSINCRONO

BALANCEAMENTO
DE CARGA

CACHE

ASYNC

CACHING

LOAD
BALANCING

Obrigado!

A thick, yellow, hand-painted style brushstroke that starts under the letter 'g' and extends to the right, ending with a slight upward curve.